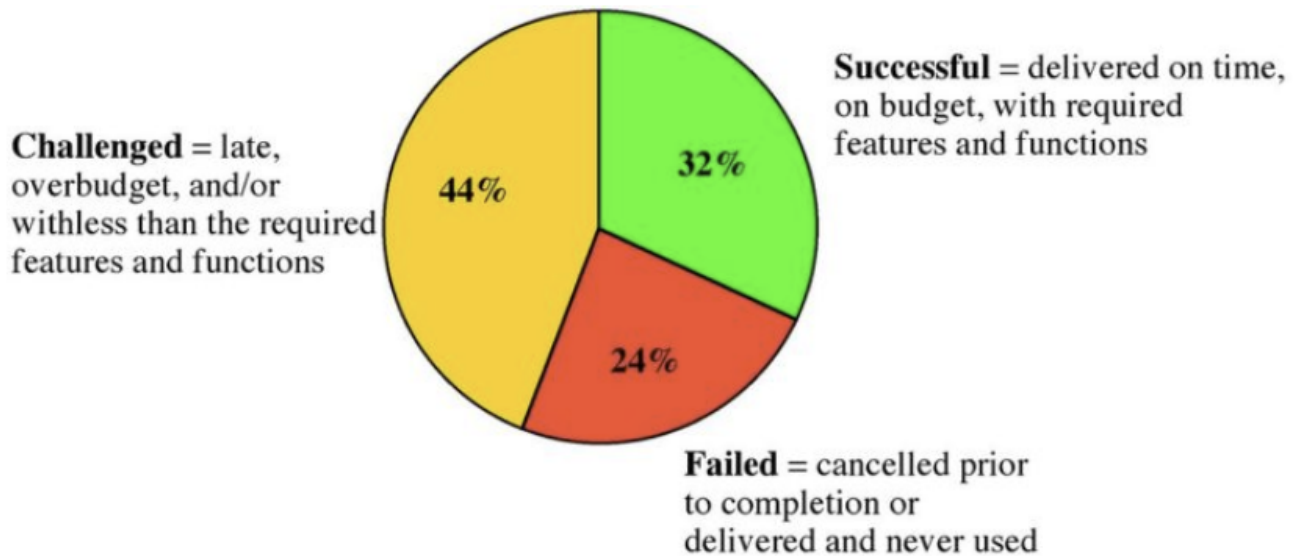


3 Projects Characteristics class

Projects Characteristics

Project success to failure ratio



Source: Summary of 2009 Standish Group CHAOS report

Projects Characteristics

Novelty:

Every project is unique. Therefore, what the project delivers has never been done before. Perhaps it has been done in a similar way before, but never exactly like this. Since we are always in uncharted territory, we must plan for known and unknown unknowns.

Time:

Projects are temporary by nature. Therefore, a significant portion of project efforts are dedicated to ensuring the project is completed on time.

Resources:

Budget for the project itself. On top of that, write down and acquire what everyone needs to perform well. Understand and address the needs of everyone involved in the project.

Success criteria:

Write down what everyone needs from the project. Understanding and addressing the needs of those affected by the project. Designing a common goal, function, or event. Once everyone understands the goal, it's much easier to keep them on the right path. Design KPIs that monitor progress based on the stated goal.

Viability:

A successful project meets or exceeds the expectations of its stakeholders. Projects are considered complete when their goals are achieved or when it is determined that the project is no longer viable. Therefore, even if 98% of a project is completed correctly and on time, it could be considered a failure if it is not finished on time or on budget.

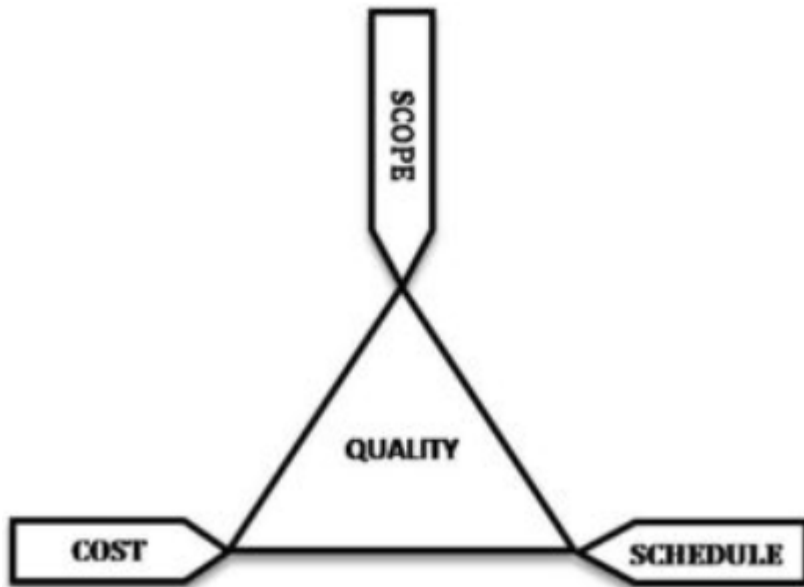
Constraints in projects

Project manager must be balancing the many competing project constraints.

- **Cost** is the budget approved for the project including all necessary expenses needed to deliver the project. Some grants have contract clauses with a "use it or lose it" approach to project funds.
- **Scope** is what the project is trying to achieve. It is the reason and the purpose of the project.
- **Quality** is a combination of the standards and criteria to which the project's products must be delivered for them to perform effectively.
- **Risk** is defined by potential external events that will have a negative impact on your project if they occur. (Alternatively a combination of the probability the event will occur and the impact on the project if the event occurs.)
- **Resources** are required to carry out the project tasks. Road blockers are the ones outside of labor.

- **Time** is defined as the time to complete the project.

Triple constraints



Triple constraints are the primary project constraints:

- Cost
- Scope
- Schedule

These are the most reactive so any changes to any one side cause a change in the other sides. The best projects have a perfectly balanced triangle.

If your sponsor wants to add functionality to the original scope, you will very likely need more money to finish the project, or if they cut the budget, you will have to reduce the quality of your scope, and if you don't get the appropriate resources to work on your project tasks, you will have to extend your schedule because the resources you have take much longer to finish the work.

Good, Quick, Cheap work constraints

- Good, Quick work won't be Cheap.
- Good, Cheap work won't be Quick.

- Quick, Cheap work won't be Good.

The process of Project Management

Project management as a process:

Project management is a process that includes planning, putting the project plan into action, and measuring progress and performance.

Bad process:

Someone's vision of what the customer wants -> Product or service development -> Implementation without customer satisfaction (because no one bothered to include the customer in the process).

Good process:

First, identify the business needs and requirements. Then, implement project management, which involves applying knowledge, skills, tools, and techniques to project activities to meet the requirements and establish a common vision across all stakeholders. Finally, define clear success criteria. For example, the goal of the Apollo Project was to land on the Moon and safely return the astronauts.

Management tensions

Poorly planned project scope can result in tensions between project manager and the people building the project.

- The generic tension is inherent in the project architecture itself. The project manager is incentivized to deliver the grandest possible project within the time and budget constraints. Those responsible for building the project, on the other hand, are incentivized to do the least amount of work possible while still delivering everything promised on time.
- One type of tension arises when the project manager demands actions and deliverables that were not included in the project plan. These demands are therefore a surprise to the people who must

carry them out. Furthermore, those additional contributions cannot be assigned to someone without a change to the project plan.

- A second type of specific tension arises when the plan assumes that certain things should be done by certain people at certain times. The project manager is incentivized to push for tasks to be completed before the deadline in order to have a time buffer. They then accumulate that buffer to deliver the project ahead of time (or use the saved time to deliver better documentation and other things that are normally deprioritized). Those responsible for building the project have an incentive to deliver by the deadline or later, if justified by external circumstances.
- The third type of tension arises when a project has KPIs for monitoring. KPIs do not move things forward; however, they are necessary for proactively addressing roadblocks and delays. Project managers have an incentive to implement as many KPIs as possible to see the situation in greater detail. Those building the project have an incentive to skip the bureaucracy and focus solely on deliverables, especially when the project timeline is at risk. Under stress, they want to cut corners by skipping the KPIs to complete the deliverables faster. As a result, the project manager loses visibility into other potential problems at a time when this visibility is more important than ever. Moreover, people who are overworked with project-related issues become demotivated by strict monitoring without justification. The project manager wants more knowledge about areas of the project where there are problems. However, all this reporting prevents people from focusing on the things that actually need to be done to move the project forward.

Adult attitude is very important in projects

To be good at project management, you must be an adult.
Nothing is more unproductive than a childish attitude.

Being an adult means:

1. You understand that your actions have consequences.
2. You understand that you will have to take responsibility for something.

The third capability stems from the first two:

3. You must be in control of your emotions.

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